

Module 1. Introduction

- ☐ What Is Data Mining?
- ☐ Data Mining: An Essential Step in Knowledge Discovery
- ☐ Diversity of Data Types for Data Mining
- ☐ Mining Various Kinds of Knowledge
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What Is Data Mining?

- ❑ We live in a world where vast amounts of data are generated constantly and rapidly
- ❑ **Data mining** is the process of discovering interesting patterns, models and other kinds of knowledge in large data sets
 - ❑ “Data mining”: a misnomer? It should be “knowledge mining from data”
 - ❑ Other terms: *Knowledge mining from data, KDD (Knowledge Discovery from Data), pattern discovery, knowledge extraction, data analytics, information harvesting*
- ❑ Data mining is a young, dynamic, and promising field
- ❑ Example: Data mining turns a large collection of data into knowledge
 - ❑ Google’s *Flu Trends* found a close relationship between the number of people who search for flu-related info. and the number of people who have flu symptoms
 - ❑ It can estimate flu activity up to two weeks faster than traditional systems

Data Mining: An Essential Step in Knowledge Discovery

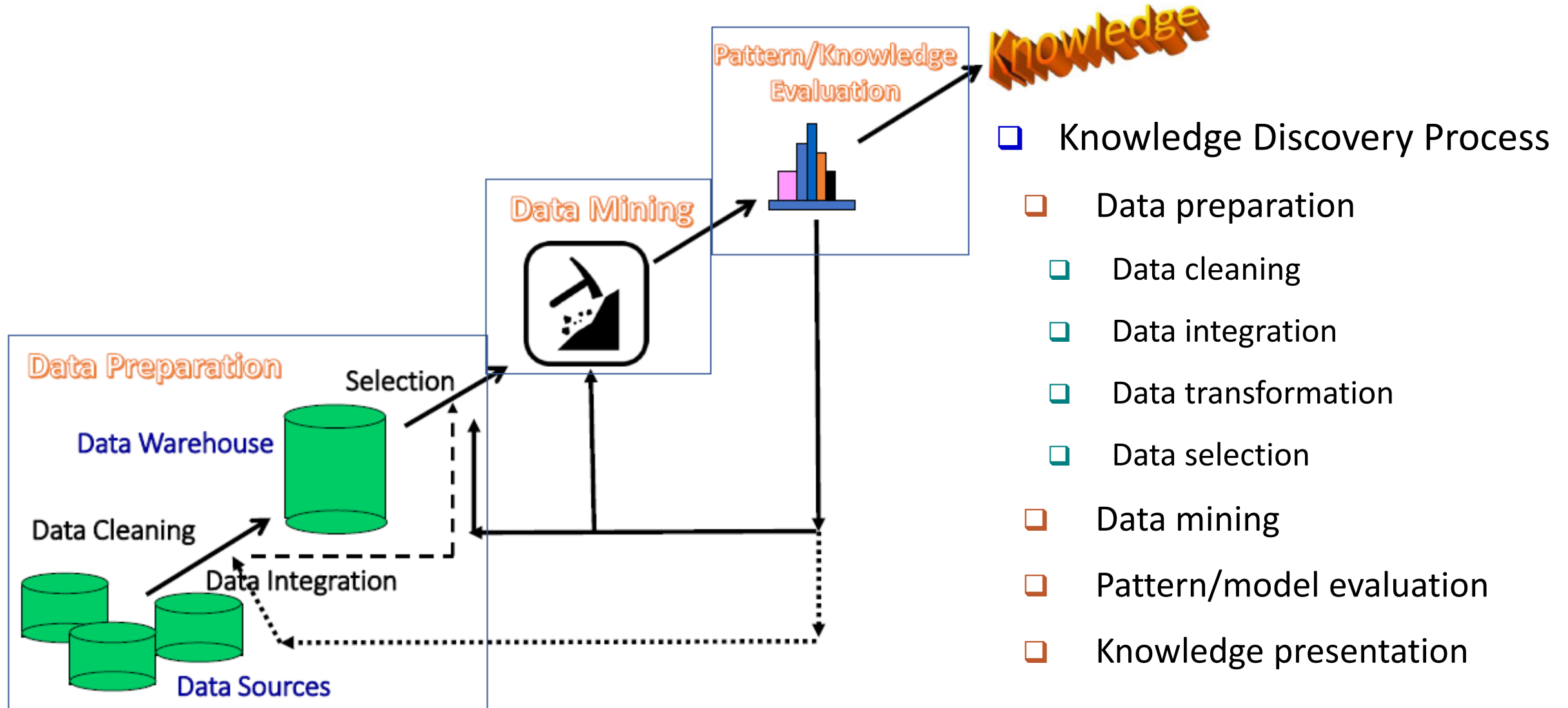


Image Source: Data Mining Concepts and Techniques Textbook (Han, Pei, Tong)